

Structuring Data in the Data Warehouse

When organizing data in a data warehouse, there are several common ways to structure it:

- 1. Simple Cumulative:** Data is accumulated over time without discarding old data.
- 2. Rolling Summary:** Data is summarized over a specific period, and older data is discarded as new data comes in.
- 3. Simple Direct:** Data is stored as it comes, without much processing or summarization.
- 4. Continuous:** Data is continuously updated and maintained in real-time.

Granularity

Granularity refers to the level of detail in your data. In a data warehouse, you can have:

- **High Granularity:** Very detailed data (e.g., data recorded every second or for each individual product).
- **Low Granularity:** Aggregated data (e.g., total monthly orders or data summarized for all products).

Key Points:

- 1. Detail and Storage:** The more detailed (higher granularity) your data is, the more storage space you will need.
- 2. Information Extraction:** Higher granularity allows for more specific and detailed reports. For example, if you store data by the second, you can generate detailed second-by-second reports.
- 3. Aggregation:** To create reports (like weekly reports), you need data that matches the required granularity (e.g., daily data to make a weekly summary).
- 4. Example:** One hour of data can be stored as:
 - 1 data point for the whole hour (low granularity)
 - 60 data points for each minute (higher granularity)
 - 3600 data points for each second (even higher granularity)
 - 3,600,000 data points for each millisecond (extremely high granularity)

Important Considerations:

- **Storage Space:** Higher granularity means more data to store, which requires more storage space.
- **Report Types:** The level of granularity determines the types of reports and analyses you can perform.

Benefits of Granularity

- 1. Reusability:** Detailed data can be used in various ways for different analyses and reports.
- 2. Flexibility:** High granularity data offers more flexibility in creating different types of reports.
- 3. Historical Data:** Granular data provides a detailed history of events and activities within the organization.
- 4. Future Requirements:** A solid foundation with detailed data can accommodate future needs and requirements, even those that are currently unknown.

In summary, understanding and choosing the right level of granularity for your data warehouse is crucial for efficient storage and meaningful data analysis.

Granularity of Data Items refers to the size of data pieces managed by concurrency control protocols, ranging from very large (entire database) to very small (individual field value).

- **Small Granularity** (e.g., single cell): Allows precise control but is complex to manage.
- **Large Granularity** (e.g., whole database): Simpler to manage but can lead to more access conflicts.

Choosing the right granularity is key to balancing efficiency and complexity in concurrent data access.

Emerging Trends & Technologies in Database Systems & Applications

Artificial Intelligence (AI)

- **AI Assistants:** Digital helpers like Siri, GoogleNow, Cortana, and Alexa use AI to perform tasks.
- **Intelligent Behavior:** AI aims to mimic human intelligence, allowing machines to learn, make decisions, and solve problems.
- **Autonomous Operation:** Machines are programmed to build knowledge and make decisions with minimal human input.

Big Data

- **Data Explosion:** With billions of Internet users and widespread use of smartphones, vast amounts of data are generated daily.
- **Volume and Complexity:** Around 2.5 quintillion bytes of data are created each day, growing with the Internet of Things (IoT).
- **Big Data:** This term describes the enormous and complex data sets produced.

Internet of Things (IoT)

- **Device Network:** Traditionally, computer networks connect devices like laptops, desktops, and smartphones through wired or wireless connections.
- **IoT Network:** IoT connects devices with embedded hardware and software, allowing them to communicate and share data.
- **Collaboration:** IoT aims to make these devices work together intelligently with minimal human intervention.

Cloud Computing

- **Internet-Based Services:** Cloud computing delivers services like software, servers, databases, and storage over the Internet.
- **Accessibility:** Users can access these services from anywhere using any device connected to the Internet.
- **Pay-Per-Use:** Cloud services are provided by companies and usually charge based on usage, similar to how we pay for electricity.
- **Scalability:** Users can run large applications or process big data without needing powerful personal computers, as long as they have an Internet connection.

In summary, AI, Big Data, IoT, and Cloud Computing are transforming how we interact with technology, manage data, and access services, making our lives more efficient and connected.

What is Big Data?

Big Data refers to extremely large and complex data sets that come from various sources like social media, sensors, and transactions. These data sets are so big that traditional tools can't easily handle or analyze them.

Why is Big Data Important?

1. Better Decisions:

- **Insights:** Big Data helps organizations understand trends and patterns, leading to smarter decisions.
- **Predictions:** It allows businesses to forecast future trends and behaviors.

2. Improved Customer Experience:

- **Personalization:** Companies can tailor products and services to individual needs, making customers happier.
- **Feedback:** Analyzing customer feedback helps improve services quickly.

3. Operational Efficiency:

- **Optimization:** Identifying inefficiencies and improving processes reduces costs.
- **Resource Management:** Efficiently using resources and reducing waste.

4. Competitive Advantage:

- **Market Insights:** Understanding market trends better than competitors provides an edge.
- **Innovation:** Finding new opportunities for products and services.

5. Risk Management:

- **Fraud Detection:** Identifying and preventing fraud.
- **Compliance:** Ensuring adherence to regulations.

6. Health and Safety:

- **Healthcare:** Improving patient care with better diagnostics and treatments.
- **Public Safety:** Anticipating and mitigating risks in emergencies.

Key Characteristics of Big Data:

1. **Volume:** The huge amount of data.
2. **Velocity:** The fast speed at which data is generated and processed.
3. **Variety:** The different types of data (structured, unstructured).
4. **Veracity:** The reliability and accuracy of the data.
5. **Value:** The useful insights and benefits that can be gained from the data.

In short, Big Data is important because it helps make better decisions, improves customer satisfaction, increases efficiency, gives a competitive edge, manages risks, and enhances health and safety.